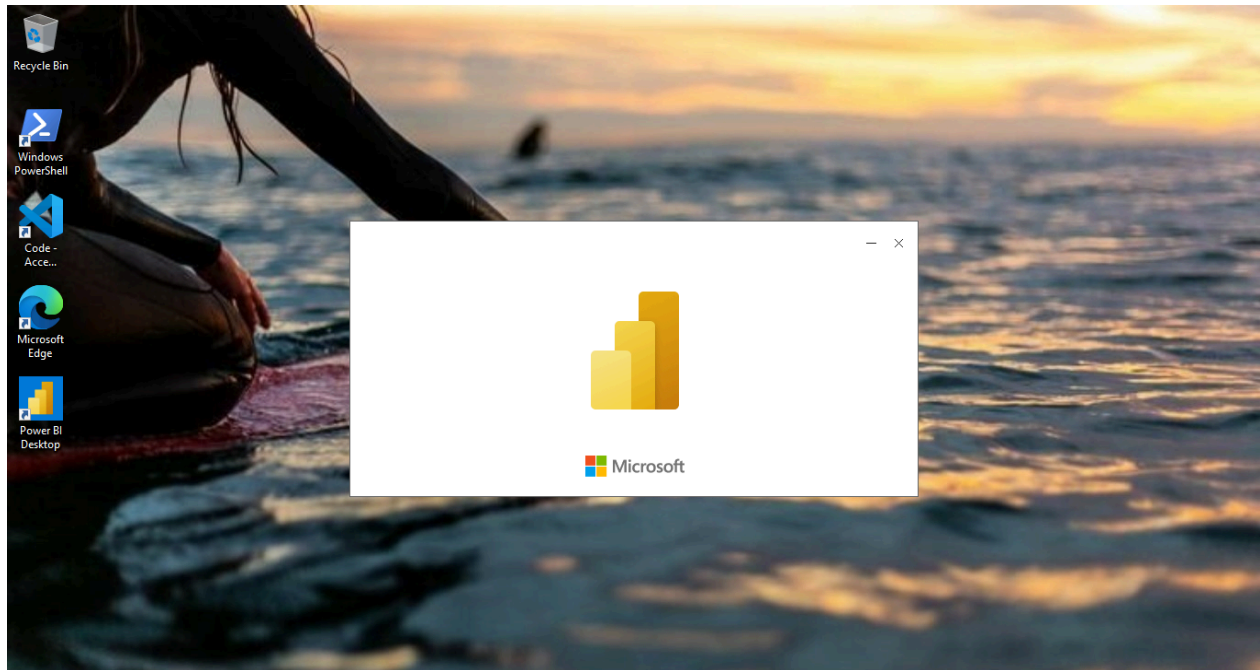


Modelado de Datos en Power BI

Práctica m46

Mayo 2026



En este proyecto desarrollé un proceso completo de preparación, limpieza y modelado de datos utilizando Power BI. El objetivo principal fue transformar un conjunto de datos crudos en un modelo estructurado que permitiera su análisis de manera eficiente y escalable.

Comencé trabajando con múltiples fuentes de datos correspondientes a distintos periodos, asegurando que todas las tablas tuvieran una estructura consistente. Durante esta fase realicé la normalización de nombres de columnas, corrección de tipos de datos y eliminación de inconsistencias que podían afectar el modelo final.

Posteriormente, consolidé la información de ventas en una sola tabla central, lo que permitió unificar el análisis y facilitar la creación de relaciones con las demás tablas dimensionales.

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Queries [8] = Table.TransformColumnTypes(#"Reordered Columns",{{"ID_Sale", Int64.Type}, {"ID_Client", Int64.Type}, {"ID_Country", Int64.Type}, {"ID_Store", Int64.Type}, {"ID_Product", Int64.Type}, {"Quantity", Int64.Type}})

	Date	ID_Sale	ID_Client	ID_Country	ID_Store	ID_Product	Quantity
1	10/24/2019	1	551	9	1	291	
2	6/17/2019	2	774	2	153	710	
3	3/4/2019	3	356	10	159	962	
4	5/15/2019	4	500	1	74	768	
5	7/14/2019	5	1	6	61	649	
6	8/13/2019	6	561	10	166	987	
7	6/13/2019	7	233	8	1	535	
8	8/10/2019	8	547	8	5	101	
9	12/6/2019	9	282	4	11	383	
10	8/22/2019	10	266	1	81	1	
11	10/15/2019	11	678	6	84	327	
12	5/11/2019	12	1	3	118	459	
13	6/7/2019	13	1	9	77	920	
14	2/3/2019	14	534	6	21	695	
15	3/25/2019	15	871	2	110	431	
16	9/8/2019	16	337	7	94	406	
17	5/21/2019	17	59	4	196	391	
18	6/10/2019	18	246	3	96	630	
19	8/26/2019	19	482	1	132	392	
20	2/7/2019	20	567	6	151	833	
21	9/20/2019	21	316	2	22	130	
22	10/11/2019	22	787	8	110	83	
23	9/27/2019	23	144	6	73	705	
24							

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Queries [8] = Table.ReorderColumns(#"Renamed Columns",{"Date", "ID_Sale", "ID_Client", "ID_Country", "ID_Store", "ID_Product", "Quantity", "Price"})

	Date	ID_Sale	ID_Client	ID_Country	ID_Store	ID_Product	Quantity
1	7/6/2020	1	124	6	4	675	
2	7/24/2020	2	498	8	70	214	
3	6/22/2020	3	752	4	76	183	
4	1/31/2020	4	808	8	62	361	
5	11/12/2020	5	323	6	170	411	
6	3/15/2020	6	461	6	95	807	
7	7/19/2020	7	759	8	130	220	
8	11/12/2020	8	693	1	29	83	
9	10/26/2020	9	559	1	107	292	
10	6/8/2020	10	346	9	89	285	
11	6/7/2020	11	240	4	43	934	
12	12/19/2020	12	6	8	182	4	
13	10/16/2020	13	422	5	8	121	
14	6/26/2020	14	680	6	49	487	
15	5/13/2020	15	155	1	180	792	
16	4/27/2020	16	822	9	78	66	
17	3/8/2020	17	225	10	84	403	
18	8/31/2020	18	224	5	161	826	
19	11/11/2020	19	55	2	74	714	
20	10/3/2020	20	367	10	9	534	
21	7/1/2020	21	536	8	64	846	
22	6/6/2020	22	245	1	103	798	
23	7/19/2020	23	454	9	121	571	
24							

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Queries [8] = Table.TransformColumnTypes(#"Reordered Columns1",{"Date", type date})

	Date	ID_Sale	ID_Client	ID_Country	ID_Store	ID_Product	Quantity
1	12/5/2021	1	94	1	102	317	
2	8/26/2021	2	30	8	16	951	
3	10/7/2021	3	83	9	38	644	
4	5/30/2021	4	82	4	82	617	
5	10/30/2021	5	26	3	35	961	
6	4/27/2021	6	58	3	142	38	
7	6/30/2021	7	45	1	57	577	
8	5/5/2021	8	72	3	136	221	
9	2/25/2021	9	5	5	41	436	
10	3/25/2021	10	44	9	128	61	
11	8/13/2021	11	92	10	174	919	
12	9/19/2021	12	29	8	27	249	
13	4/2/2021	13	11	5	35	887	
14	9/19/2021	14	88	10	138	452	
15	7/25/2021	15	2	6	172	980	
16	11/27/2021	16	66	2	177	733	
17	6/9/2021	17	97	1	146	163	
18	11/14/2021	18	8	4	128	222	
19	6/13/2021	19	93	1	4	523	
20	7/18/2021	20	89	4	150	37	
21	12/17/2021	21	47	10	21	316	
22	2/4/2021	22	83	1	143	457	
23	8/24/2021	23	97	7	83	265	
24							

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Queries [8] = Table.Distinct(#"Removed Duplicates", {"Product ID"})

	Product ID	Category	Sub-Category	Product Name	Active
1	999	Office Supplies	Appliances	Cuisinart Stove, Black	1
2	998	Furniture	Tables	Hon Training Table, Adjustable Height	1
3	997	Furniture	Bookcases	Safco Library with Doors, Metal	1
4	996	Office Supplies	Binders	Ibico Ibimaster 300 Manual Binding System	1
5	995	Furniture	Bookcases	Sauder Floating Shelf Set, Metal	1
6	994	Office Supplies	Appliances	Breville Refrigerator, Red	1
7	993	Office Supplies	Appliances	Breville Stove, Red	1
8	992	Technology	Copiers	Brother Wireless Fax, Digital	1
9	991	Technology	Phones	Samsung Galaxy Note 3	1
10	990	Office Supplies	Appliances	Kensington 7 Outlet MasterPiece HOMEOFFICE Power Control Center	1
11	989	Office Supplies	Storage	Fellowes Lockers, Blue	1
12	988	Office Supplies	Storage	Fellowes File Cart, Industrial	1
13	987	Furniture	Chairs	Harbour Creations Swivel Stool, Black	1
14	986	Office Supplies	Appliances	Breville Stove, White	1
15	985	Office Supplies	Storage	Smead Lockers, Blue	1
16	984	Furniture	Bookcases	Ikea Library with Doors, Traditional	1
17	983	Technology	Copiers	Sharp Copy Machine, Laser	1
18	982	Furniture	Bookcases	Safco Floating Shelf Set, Pine	1
19	981	Office Supplies	Appliances	Hamilton Beach Stove, Silver	1
20	980	Technology	Phones	Mitel MiVoice 5330e IP Phone	1
21	979	Office Supplies	Storage	Dual Level, Single-Width Filing Carts	1
22	978	Technology	Accessories	Belkin Router, Ergonomic	1
23	977	Furniture	Tables	Lesro Conference Table, with Bottom Storage	1
24	976	Technology	Machines	Okidata Inkjet, Wireless	1

5 COLUMNS, 999 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 11:09 AM

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Queries [8]

- sales_2019_DN
- sales_2020_DN
- sales_2021_DN
- t_customer_DN
- t_store_DN
- t_countries_DN
- t_product
- sales_all_DN

Table.Distinct("#Changed Type2")

ID_Country	Country
1	US
2	CANADA
3	MEXICO
4	BRAZIL
5	SPAIN
6	COLOMBIA
7	CHILE
8	PERU
9	PORTUGAL
10	FRANCE

2 COLUMNS, 10 ROWS Column profiling based on top 1000 rows

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Queries [8]

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Table.TransformColumnTypes("#Removed Duplicates",{"ID_Store", Int64.Type}, {"Store_Name", type text}, {"Address", type text}, {"Format", type text}, {"Status", type text})

ID_Store	Store_Name	Address	Format	Status
1	Huels-Robel	1029 Petterle Trail	AAA	Open
2	Smith-Lowe	8 Lake View Trail	AAA	Open
3	Dickens-Oberbrunner	86 Chive Way	AAA	Open
4	Macejkovic, Hermann and Bosco	4409 Novick Junction	AAA	Open
5	Swift Group	2 Macpherson Pass	AAA	Open
6	Rippin LLC	30 Badeau Parkway	AAA	Open
7	Effertz-Ullrich	32331 Merrick Trail	AAA	Open
8	Welch, Hartmann and Durgan	78 Magdeline Way	AAA	Open
9	Morar-Flatley	7 Burning Wood Park	AAA	Open
10	Kiehn Group	63239 Hovde Junction	AAA	Open
11	Hamill and Sons	4420 Schmedeman Parkway	AAA	Open
12	Auer Group	3 Red Cloud Center	AAA	Open
13	Schumm Inc	2815 Buell Court	AAA	Open
14	Toy Inc	2060 Toban Way	AAA	Open
15	Douglas Group	18 Linden Terrace	AAA	Open
16	Nikolaus, Harvey and Waelchi	271 Village Green Lane	AAA	Open
17	Sawayn-Reichert	64 Merry Terrace	AAA	Open
18	Stamm, Kuhic and Bosco	740 Monica Hill	AAA	Open
19	Schuster, Schulist and Jaskolski	29133 Fairfield Junction	AAA	Open
20	Kirlin Inc	103 Ohio Drive	AAA	Open
21	Wuckert, Kozey and Collier	5257 Autumn Leaf Road	AAA	Open
22	VonRueden, Hane and Bernier	1680 Northland Trail	AAA	Open

5 COLUMNS, 200 ROWS Column profiling based on top 1000 rows

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Construcción del Modelo de Datos

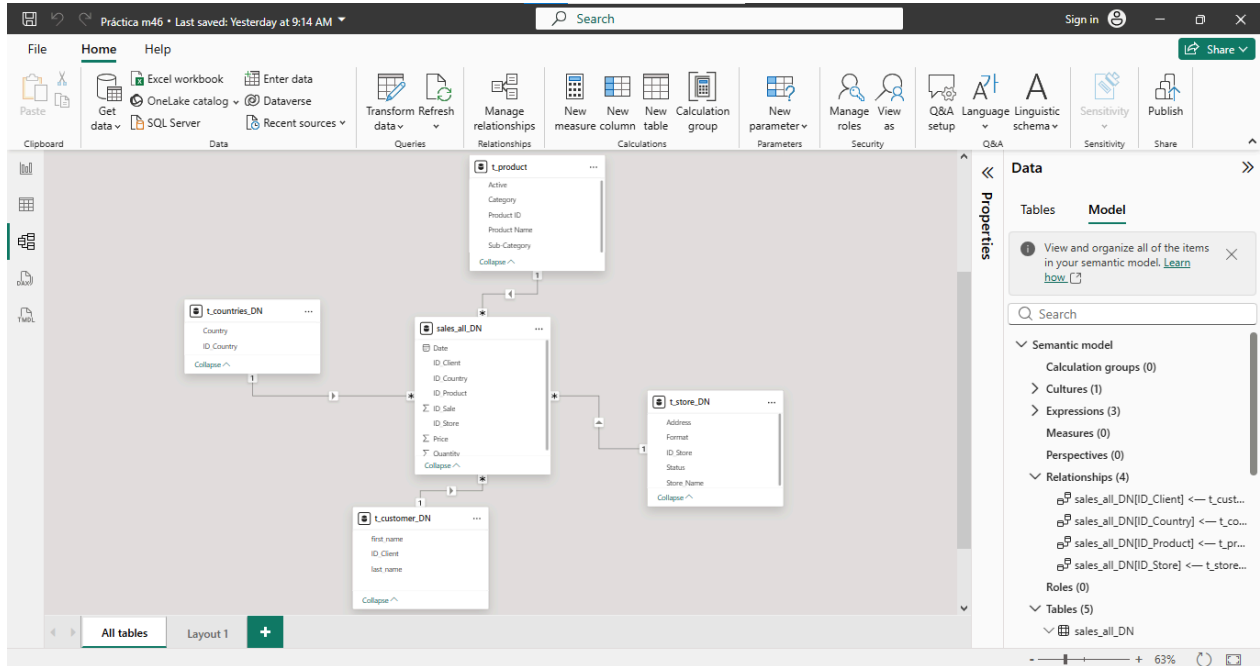
Una vez finalizada la limpieza de los datos, procedí a estructurar el modelo utilizando un enfoque tipo estrella. Definí una tabla de hechos central que concentra las transacciones de ventas y la relacioné con diferentes tablas dimensionales como productos, clientes, tiendas y países.

Este enfoque permite optimizar el rendimiento del modelo y facilita la creación de análisis más claros, ya que cada dimensión aporta contexto específico a los datos de ventas.

Durante este proceso validé que todas las relaciones fueran correctas, asegurando la integridad referencial entre las tablas y evitando relaciones duplicadas o inconsistentes.

Conclusión del Modelo

El resultado final es un modelo de datos estructurado, eficiente y listo para ser utilizado en la creación de reportes y dashboards. La implementación del esquema en estrella garantiza una navegación clara de los datos y una base sólida para futuros análisis.



Reflexión Final

A través de este proyecto comprendí la importancia de la preparación de datos como base fundamental en cualquier proceso de análisis. La calidad del modelo depende directamente de la limpieza y organización previa de la información.

También reforcé el uso de relaciones entre tablas y la aplicación de modelos dimensionales, lo que permite transformar datos aislados en información útil para la toma de decisiones.

Este ejercicio consolidó mi capacidad para estructurar datos de manera profesional, aplicando buenas prácticas utilizadas en entornos reales de análisis.

RobertsScience Data Analytics Consulting
Convertimos datos en resultados